

MICHAEL KINYUNYU

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience developing Python-based applications, REST API integrations, database solutions, automation scripts, data processing systems, and containerized engineering workflows. Experienced with Python, SQL, PostgreSQL, Git, Docker, Linux environments, AWS, FastAPI, PySpark, Apache Airflow, and Apache Kafka. Built and deployed practical software and data-intensive systems involving API integration, backend processing, database operations, automation, monitoring, and cloud infrastructure. Strong foundation in software development lifecycle activities including design, implementation, testing, debugging, documentation, deployment, and system troubleshooting.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML, CSS

Backend & APIs: FastAPI, REST APIs, API Integration, Backend Development, JSON, API Data Processing

Databases: PostgreSQL, SQL, Relational Database Systems, Data Modeling, Database Queries

Linux & Automation: Linux, Bash/Shell Scripting, System Troubleshooting, Python Automation, Command-Line Tools

Cloud & Infrastructure: AWS S3, AWS Glue, Amazon Athena, Databricks

Containers & DevOps: Docker, Git, GitHub, CI/CD Fundamentals, Containerized Applications

Data & Distributed Systems: PySpark, Apache Kafka, Apache Airflow, ETL/ELT, Batch Processing, Streaming Pipelines

Monitoring & Analytics: Grafana, Amazon QuickSight, Tableau, Crystal Reports

Software Engineering: SDLC, Debugging, Testing, Technical Documentation, Version Control, System Integration

PROJECTS

1. SMS Intelligent Monitoring System

- **Tech Stack:** Python + FastAPI + REST API + Docker + ML + backend development Git | [github](#)
- I built and deployed a Python/FastAPI backend application that exposes machine-learning inference through REST APIs and is containerized with Docker

2. OmniStream Analytics – Multi-Layered E-Commerce Growth Pipeline

- **Tech Stack:** Databricks, PySpark, AWS S3, Python, Streaming/Batch Pipelines | [github link](#)
- Designed a Bronze → Silver → Gold e-commerce data platform using Databricks, PySpark, AWS S3, and Python for transactional and event-driven data.
- Built automated data ingestion, quality validation, schema reconciliation, and monitoring workflows to improve reliability across high-volume pipelines.
- Developed curated Gold-layer datasets for customer purchasing behavior, product performance, revenue, and business analytics

3. Healthcare Cancer Data Intelligence Pipeline

- **Tech Stack:** AWS S3, AWS Glue, PySpark, Amazon Athena, Amazon QuickSight | [Github link](#)

- Built Python REST API ingestion workflows to extract source data and persist raw datasets into AWS S3.
- Developed automated PySpark ETL pipelines to clean, normalize, validate, and schema-enforce inconsistent source data.
- Created curated, partition-optimized Parquet datasets and AWS Glue catalog structures for governed querying through Athena and QuickSight

EXPERIENCE

Software Engineering Field

Aug 2025 – Oct 2025

Tanzania Telecommunication Corporation Limited

Iringa, Tanzania

- Developed SQL queries supporting operational reporting and production database activities.
- Built Python scripts to automate recurring system-maintenance tasks and reduce manual operations.
- Participated in software development activities spanning design, implementation, testing, documentation, and deployment.

Software Engineering Field

Jul 2024 – Sep 2024

TEKU

Mbeya, Tanzania

- Developed SQL queries for application and operational data requirements.
- Built web interfaces using HTML, CSS, and JavaScript.
- Automated recurring operational tasks using Python.
- Investigated and resolved application functionality and debugging issue

EDUCATION

Bachelor of Software Engineering

June 2026

Ruaha Catholic University

Iringa, Tanzania